Combination circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Reduce a circuit by identifying genuine series and parallel groups.

SUCCESS CRITERIA - I can...

1. Identify nodes before grouping resistors.
2. Reduce one group at a time.
3. Check current and voltage conservation in the original circuit.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 Ω resistor carries 5 A. Find the potential difference across it.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $R_{series} = R_1 + R_2$; $1/R_{parallel} = 1/R_1 + 1/R_2$; $V = IR$. Reduce one known group at a time. Check nodes before calling components series or parallel.

Worked reasoning: A 2 ohm resistor is in series with two 4 ohm resistors in parallel. The parallel pair equals 2 ohms; total 4 ohms. A 12 V supply gives 3 A total.

Your turn: Draw the network and label the two series sections before splitting the branch current.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

A 2 Ω resistor is in series with two parallel 6 Ω resistors. Draw the arrangement and circle the parallel group.

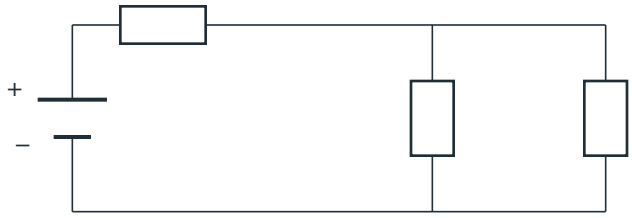


Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A 10 V battery powers the network in question 3: a 2 Ω resistor in series with two parallel 6 Ω resistors. You may use the supplied parallel equivalent, 3 Ω. Label each section, find total resistance and calculate source current.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

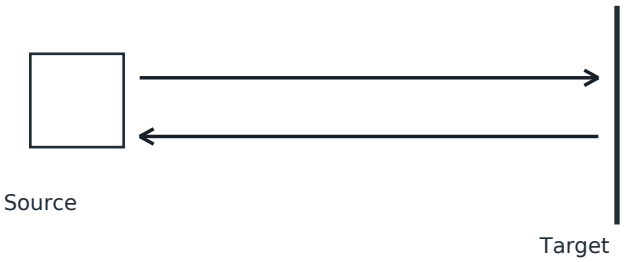
05 Explain your reasoning

A student adds all three resistor values because they are drawn side by side. What should determine the grouping instead?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A radar echo’s complete outward-and-return path is 6000 m. Find the target range and explain the division. Label the outgoing and returning paths.



Marzano check - circle: 1 2 3

07 Exit check

If 2 A reaches two equal parallel branches, find each branch current. Label the circuit and show the current path or paths.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Replace the parallel pair by two 8 ohm resistors. Find the new equivalent resistance and total current on 12 V. Label the circuit and show the current path or paths.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3